

2023 年春大学物理 IA 期末考试试题（回忆版）答案

精简公式版；默认按题图与大学物理常用记号书写。

一、选择题

题号	1	2	3	4	5	6	7	8	9	10
答案	C	C	A	A	A	C	A	B	B	D

1. 速率减小： $\mathbf{a} = \mathbf{a}_\tau + \mathbf{a}_n$ ，切向分量反向、法向分量指向曲率中心，合成向下。

2. $T \cos \alpha = mg$, $T \sin \alpha = m\omega^2 l \sin \alpha$, 故

$$T = \frac{mg}{\cos \alpha}, \quad \omega = \sqrt{\frac{g}{l \cos \alpha}}.$$

3. 仅 (1) 正确。

4. $\Delta t = \gamma \Delta t'$, $\gamma = 5/3$,

$$v = c \sqrt{1 - \frac{1}{\gamma^2}} = 0.8c.$$

5. $r < \rho < R$ 时 $E = \frac{q}{4\pi\epsilon_0\rho^2}$ 。

6. 球壳内 $E_Q = 0$, $V_Q = \frac{Q}{4\pi\epsilon_0 R}$ 。

7. 内壳接地后 $0 = \frac{q}{4\pi\epsilon_0 R_1} + \frac{q_2}{4\pi\epsilon_0 R_2}$,

$$q = -\frac{R_1}{R_2} q_2, \quad q + q_2 = q_2 \left(1 - \frac{R_1}{R_2}\right) > 0.$$

8. $\frac{\mu_0 I_1}{2\pi(0.50)} = \frac{\mu_0 I_2}{2\pi(1.50)}$, 故 $I_2 = 18 \text{ A}$, 方向垂直流入纸面。

9. 圆电流与回路共面, $\mathbf{H} \perp d\mathbf{l}$, 故 $\oint_L \mathbf{H} \cdot d\mathbf{l} = 0$ 。

10. $B = \frac{\mu_0 I}{2R}$,

$$w = \frac{B^2}{2\mu_0} = \frac{\mu_0 I^2}{8R^2}.$$

二、填空题

1. 雨滴竖直速度 u 满足 $\tan 30^\circ = 20/u$, 相对汽车速率

$$v_{\text{rel}} = \sqrt{20^2 + u^2} = 40 \text{ m s}^{-1}.$$

2. 整杆 $a = F/m$, $m_{BC} = m/5$,

$$N = m_{BC}a = \frac{F}{5}.$$

3. $\alpha = \Delta\omega/\Delta t = 5/10 = 0.5 \text{ rad s}^{-2}$,

$$J = \frac{M}{\alpha} = 40 \text{ kg m}^2.$$

4. $\gamma = (1 - 0.8^2)^{-1/2} = 5/3$,

$$K = (\gamma - 1)m_0c^2 = \frac{2}{3}(0.51 \text{ MeV}) = 0.34 \text{ MeV}.$$

5. 按相对论质量口径:

$$\rho' = \frac{\gamma m}{a^3/\gamma} = \frac{\gamma^2 m}{a^3} = \frac{4m}{3a^3} \quad (v = 0.5c).$$

6. 小孔等效为无限大带电平板减去圆盘, $x = R/\sqrt{3}$:

$$E = \frac{\sigma}{2\varepsilon_0} \frac{x}{\sqrt{x^2 + R^2}} = \frac{\sigma}{4\varepsilon_0},$$

方向沿轴线离开带正电平板。

7. 两壳间 $E = \frac{q_1}{4\pi\varepsilon_0\varepsilon_r r^2}$,

$$W = \int_{R_1}^{R_2} \frac{1}{2} \varepsilon_0 \varepsilon_r E^2 4\pi r^2 dr = \frac{q_1^2}{8\pi\varepsilon_0\varepsilon_r} \left(\frac{1}{R_1} - \frac{1}{R_2} \right).$$

8.

$$E = \frac{U}{d} = \frac{UC}{\varepsilon_0\varepsilon_r S} = \frac{50 \times 100 \times 10^{-12}}{8.85 \times 10^{-12} \times 6 \times 100 \times 10^{-4}} \approx 9.4 \times 10^3 \text{ V m}^{-1}.$$

9. $r = \frac{mv}{qB}$,

$$\Phi_m = B\pi r^2 = \frac{\pi m^2 v^2}{q^2 B}.$$

10. 按题图中 $\mathbf{B}, \mathbf{v}, \overline{ab}$ 均在纸面内:

$$\mathcal{E}_{ab} = \int_a^b (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = 0.$$

若原题意为 $\mathbf{B}$ 垂直纸面, 则 $\mathcal{E} = Blv \sin \alpha$ 。

三、计算题

1. 子弹嵌入杆

$$J_{\text{rod}} = \frac{1}{3}ml^2 = 0.50 \text{ kg m}^2, \quad J_{\text{tot}} = J_{\text{rod}} + m'l^2 = 0.52 \text{ kg m}^2.$$

碰撞过程对转轴角动量守恒:

$$m'vl = J_{\text{tot}}\omega, \quad \omega = \frac{0.020 \times 400 \times 1.0}{0.52} \approx 15.4 \text{ rad s}^{-1}.$$

摩擦力矩制动:

$$\alpha = \frac{M_f}{J_{\text{tot}}}, \quad t = \frac{\omega}{\alpha} = \frac{J_{\text{tot}}\omega}{M_f} = \frac{m'vl}{M_f} = \frac{8.0}{4.0} = 2.0 \text{ s}.$$

2. 同轴带电圆柱面外电势

$$E(\rho) = \begin{cases} \frac{\lambda}{2\pi\epsilon_0\rho}, & R_1 < \rho < R_2, \\ 0, & \rho > R_2. \end{cases}$$

取 $V(R_1) = 0$:

$$V(R_2) - V(R_1) = - \int_{R_1}^{R_2} \frac{\lambda}{2\pi\epsilon_0\rho} d\rho = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}.$$

故对 $r > R_2$,

$$V(r) = - \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}.$$

3. 偏心空腔载流圆柱

电流密度

$$j = \frac{I}{\pi(R_1^2 - R_2^2)}.$$

用“完整半径 R_1 圆柱电流 $+j$ ”叠加“空腔半径 R_2 圆柱电流 $-j$ ”:

$$B_{\text{大轴}} = \frac{\mu_0 j R_2^2}{2b} = \boxed{\frac{\mu_0 I R_2^2}{2\pi b(R_1^2 - R_2^2)}}.$$

$$B_{\text{空腔轴}} = \frac{\mu_0 j b}{2} = \boxed{\frac{\mu_0 I b}{2\pi(R_1^2 - R_2^2)}}.$$

4. 两密绕长螺线管

长螺线管近似:

$$B_i = \mu \frac{N_i}{l} I_i, \quad \Phi_i = BS.$$

自感:

$$\boxed{L_1 = \mu \frac{N_1^2 S}{l}}, \quad \boxed{L_2 = \mu \frac{N_2^2 S}{l}}.$$

互感:

$$\boxed{M = \mu \frac{N_1 N_2 S}{l}}.$$

同向绕制时

$$\boxed{M = \sqrt{L_1 L_2}}.$$